



THESIS ASSIGNMENT

Name and Surname: Bc. Jitka Muravská
Study programme: Computer Science (Single degree study, master II. deg., full time form)
Field of Study: Computer Science
Type of Thesis: Diploma Thesis
Language of Thesis: English
Secondary language: Slovak

Title: Permuting Matrix Columns for Improved Block-Sparse Pruning

Annotation: In currently used neural networks, a significant portion of computational time is dedicated to matrix multiplications involving weight matrices and activations. Research indicates the feasibility of pruning substantial portions of weight matrices to zero without a significant sacrifice in accuracy. Despite this, practical speed-ups remain elusive, as the multiplication of sparse matrices is inefficient on current hardware. Block sparsity offers zeros in small blocks of matrices and ones, and offers a much friendlier hardware profile.

Ji et al. [1] showed that it is possible to permute the weight matrix first and achieve better accuracy for the same sparsity. The goal of this thesis is to explore and improve matrix permutation algorithms.

[1] Ji, Yu, et al. "Tetris: Tile-matching the tremendous irregular sparsity." Advances in neural information processing systems 31 (2018).

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Assigned: 01.12.2023

Approved: 30.04.2026
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